

DATA607 - Assignment 2:

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Assignment Details

Assignment: SQL and R – Movie Ratings Objective Collect simple movie-rating data, store it in a SQL database, and analyze it in R.

Implementation Notes: PostgreSQL is the recommended (and supported) relational database for this assignment. If you are unable to install and configure PostgreSQL in your environment, you may find SQLite easier to use. Students experienced with relational databases may use any local or cloud-based database for their assignment. NEVER include passwords in your code. For this assignment, the ability to reproduce your work in my environment is not required, but you need to provide all of the necessary code, include the code used to create and populate tables. **Task Description**

Select six recent popular movies (or television episodes or books or songs or ...)

Ask at least five people to rate each movie they have seen on a 1–5 scale.

Store the collected ratings in a SQL database of your choice.

Load the data from SQL into R as a dataframe.

##Implementation Notes

The assignment is intentionally open-ended; multiple approaches are acceptable.

You may:

Query the SQL database directly from R, or

Export SQL data to a .csv file and load it into R.

If you use a GUI to create or populate tables, you must also provide the SQL code needed to recreate them.

##Reproducibility

Your solution should be reproducible assuming the same SQL server and R environment.

You may remove or mask database passwords.

Full reproducibility of survey responses is not required.

##Missing Data

You must demonstrate a reasonable strategy for handling missing ratings, since not all participants will have seen all movies.

Optional Enhancements You may improve your solution by:

Using survey software to collect ratings

Managing database credentials securely

Designing a normalized database schema

Exploring whether standardizing ratings is beneficial

Submission Requirements

Post all code (SQL, R, Quarto) in a GitHub repository

Submit the GitHub link

RPubs publication is not required

Group work is allowed, but each student must submit individually and list collaborators

Professional Expectation Treat this as portfolio-quality work suitable for a job interview.

Approach

Type out approach here